AI-Based Design Change Impact Analysis System

Objective

The main goal of this project is to build an AI-based system that can take a design change as input and automatically:

- Identify which systems will be affected
- Estimate the cost of the change
- Validate compliance rules
- Provide insights using historical data

This helps in reducing manual analysis effort and improves decision making speed.

Use Cases

- Analyze the impact of a design change across multiple interconnected systems
- Estimate cost based on the type and magnitude of change
- Automatically validate compliance and class rules
- Use historical data to support better and faster decision-making

Concept Overview

This system follows a layered and scalable architecture:

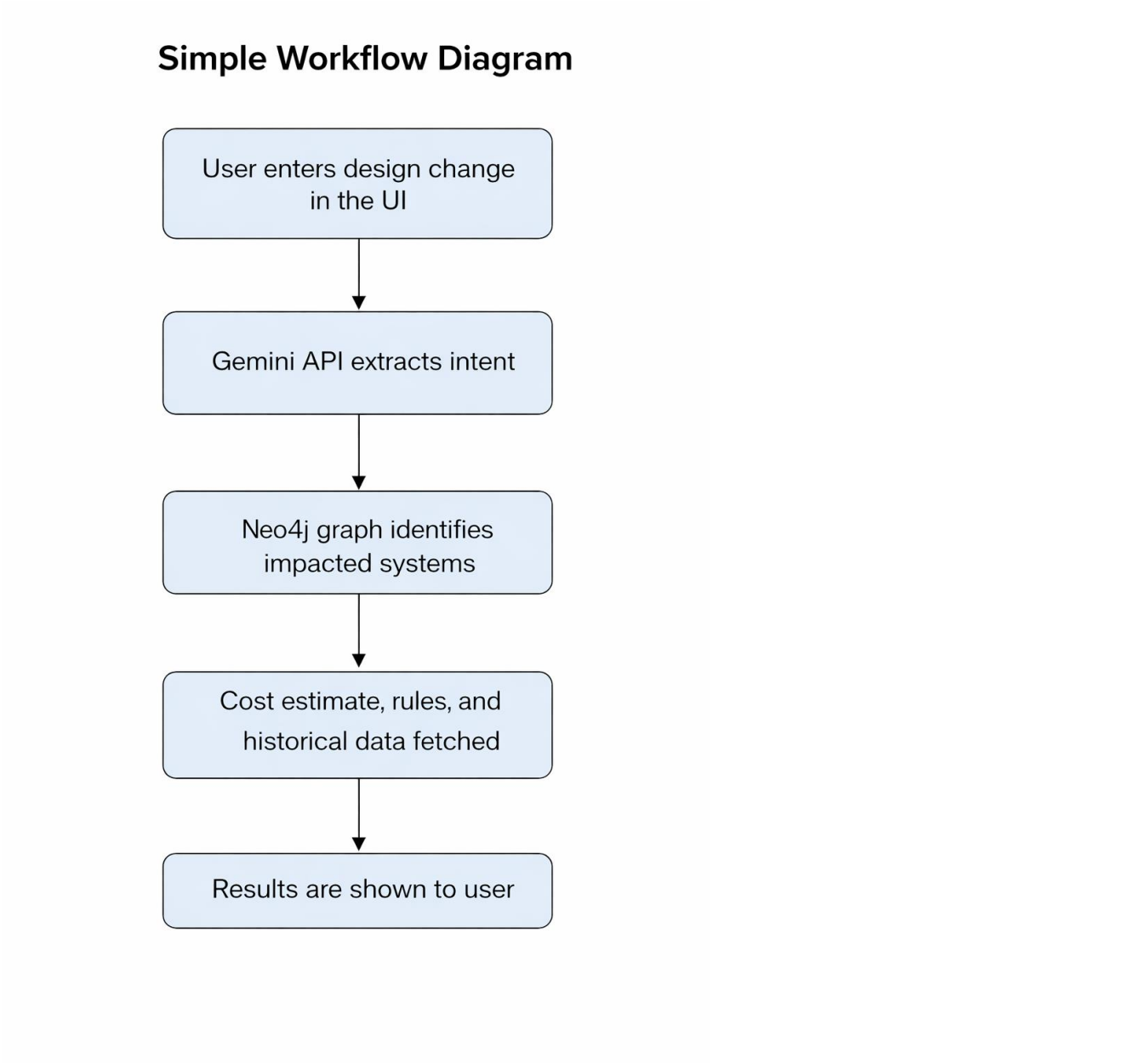
- **Frontend (UI) =** User interacts and views results
- **Backend (FastAPI) =** Handles business logic and API processing
- **Database (Neo4j) =** Stores and manages system relationships in graph form

For AI processing, **Google Gemini API** is used for natural language understanding. It converts user input into structured intent, which drives graph queries and cost estimation

Process / Workflow

- User enters a design change in the UI
- Backend sends input to Gemini API for intent extraction
- Extracted intent is converted into structured parameters

- Neo4j database is used to find how a change in one system affects other connected systems
- Cost model calculates estimates (P10, P50, P90)
- Compliance rules and historical data are retrieved
- Final response is displayed on the UI



Work Done

- Understood and analyzed the complete PoC architecture
- Designed and implemented REST APIs using FastAPI
- Used Neo4j to represent and manage system relationships.
- Built frontend using React and Tailwind CSS
- Integrated Google Gemini API for intelligent intent extraction
- Connected Gemini API with backend to process user queries
- Developed an end-to-end working integrating AI, backend, database, and UI
- Implemented cost estimation and rule validation
- Resolved API communication issues, including CORS
- Tested the system with multiple real-world-like scenarios

Advantages

- Automates impact analysis and reduces manual effort
- Provides fast, structured and scalable cost estimation
- Uses graph-based modeling for accurate dependency analysis
- Improves decision-making using historical insights
- Designed with a scalable architecture for future enhancements

Limitations

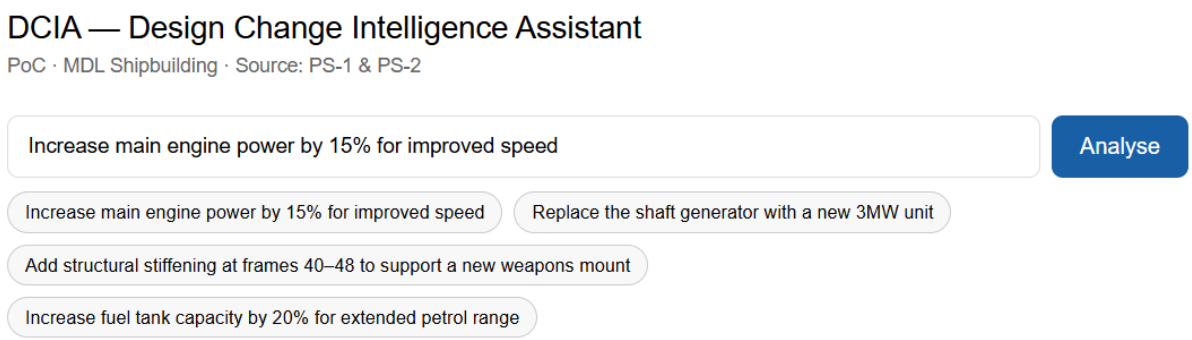
- Currently uses limited AI capability for intent detection, which may not handle highly complex inputs
- Working on sample data, which may not fully represent real-world scenarios
- Accuracy can be improved with real-world datasets and advanced AI/ML models

Next Steps

- Test with real-world data and more complex scenarios
- Enhance UI with advanced visualizations and dashboards
- Improve backend logic and AI processing for higher accuracy
- Extend system for production-level deployment

Sample Output Explanation

When a user enters a change like **“Increase main engine power by 15% for improved speed”**, the system gives:



- Impacted systems: 6
- Rules triggered: 4
- Cost estimate:
    - P10 → lower estimate
    - P50 → average estimate
    - P90 → higher estimate

It also gives analysis and conclusions about feasibility and risk.

Detailed Impact

The system displays affected systems along with severity levels:

- Hull Structure → High impact
- Electrical Power → Medium
- Shafting & Gearbox → High
- Fuel System → Medium

- Others → Low

Also shows:

- Cost graph (P10, P50, P90)
- Rules triggered
- Historical similar case

Impact assessment

Design Change Impact Assessment:

User Request:
Increase main engine power by 15% for improved speed

Key Findings:
- Total impacted systems: 6
- Class rules triggered: 4
- Historical analogues found: 1

Cost Estimate (USD):
- P10 (Optimistic): $4,947,188
- P50 (Most Likely): $6,596,250
- P90 (Pessimistic): $8,904,938

Analysis:
The proposed change has cascading impacts across multiple ship systems.
Critical dependencies across propulsion, structure, and electrical systems
should be carefully reviewed before implementation.

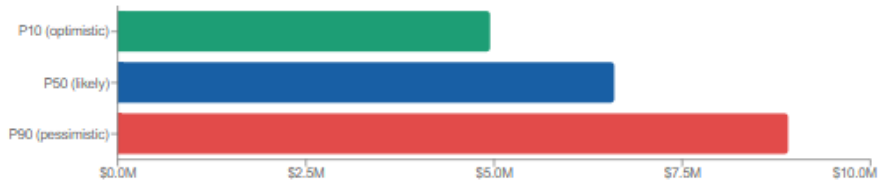
Conclusion:
The design change is feasible with moderate cost and manageable risk.
Further validation using detailed engineering analysis is recommended.

(Note: This is a mock AI-generated report for PoC demonstration.)
Source: Neo4j graph traversal + LLM synthesis · Confidence: MEDIUM

Affected systems (6)

System	Discipline	Hops from origin	Severity	Reason
Hull Structure	Structure	1	HIGH	—
Electrical Power	Electrical	1	MEDIUM	—
Shafting & Gearbox	Propulsion	1	HIGH	—
Fuel System	Outfitting	1	MEDIUM	—
Outfitting / HVAC	Outfitting	2	LOW	—
Weapons Systems	Weapons	2	LOW	—

Cost estimate (USD)



PoC parametric model — replace with MDL historical data regression in production

Class rule flags (4)

LR — Pt3 Ch2 Sec5.3
Main engine power-to-weight ratio limits for naval vessels
Triggered by: Main Propulsion

DNV — RU-SHIP Pt4 Ch2
Vibration limits for hull structure linked to propulsion
Triggered by: Hull Structure

DNV — RU-SHIP Pt4 Ch6
Electrical load balance requirements after machinery change
Triggered by: Electrical Power

LR — Pt3 Ch4 Sec2.1
Shaft alignment tolerances post power upgrade
Triggered by: Shafting & Gearbox

Historical analogues

HC-2022-014 — Engine upgrade from MTU 3600 to MTU 4000 on OPV class
Power increase 12%. Required shaft re-alignment, electrical load re-calc, hull stiffening at frames 42-46.
Actual cost: $3,200,000 · Schedule delay: 28 days

Graph Model

The Neo4j graph represents how systems are interconnected:

- Nodes represent systems (engine, fuel, etc.)
- Edges represent relationships between systems
- Helps in identifying cascading impacts across systems

Conclusion

The system is currently in the Proof of Concept (PoC) stage.
It shows strong potential for real-world use with better data and advanced AI models.
It can be further improved and scaled for production.